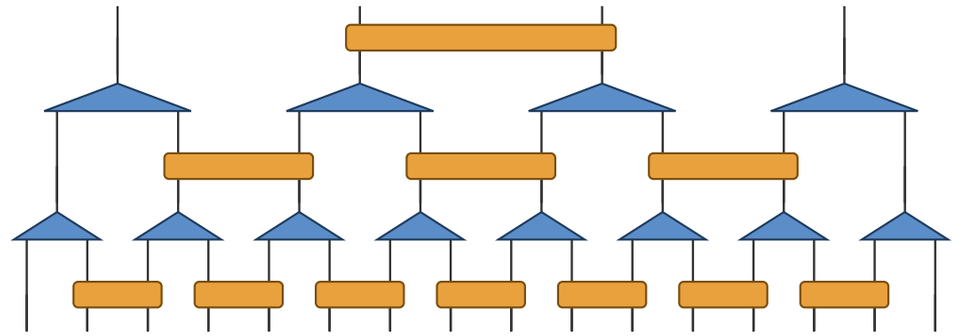
Tensor networks methods for low-dimensional quantum systems



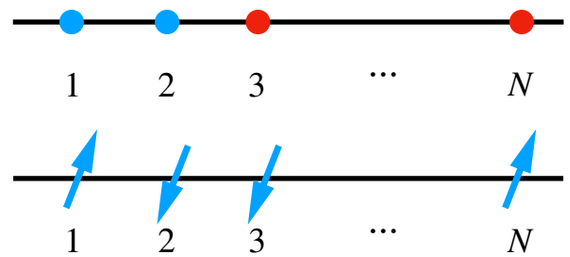
Lecture 2: Entanglement entropy

UNIVERSITY OF WARSAW

WYDZIAŁ FIZYKI

Last week

1d models on a lattice **spins, fermions, bosons**



transformations

- **spins <-> hard-core bosons**
- **hard-core boson <-> fermions (Jordan-Wigner)**
- **spins <-> bosons (Holstein-Primakoff)**

We will mostly talk about spin chains:

- **theory of quantum magnetism**
- **quantum computations (qubit = spin 1/2)**

XYZ spin chain:
the most general neareast neighbour

$$\hat{H} = \sum_{j=1}^{N} \left(J_x s_j^x s_{j+1}^x + J_y s_j^y s_{j+1}^y + J_z s_j^z s_{j+1}^z \right) + h \sum_{j=1}^{N} s_j^z$$

$J_x = J_y$ **- XXZ spin chain - solvable by Bethe Ansatz**

$J_z = 0$ **- XY spin chain, a generalization of the Ising chain - solvable by Jordan-Wigner + Bogoliubov**

Questions and Strategies

Questions:

1) **ground state phase diagram - quantum phase transitions**
2) **finite temperature - closer to experiments**
3) **non-equilibrium dynamics - foundations of stat. mech.**

Strategies:

1) Exact solutions - rare and important - benchmark point for approximate methods

2) Numerical diagonalization - numerically exact - limited to small system sizes

3) Quantum Monte Carlo - variational wavefunction - good for bosonic systems and finite T

4) Tensor Networks - matrix product state - great for 1d ground-states

Today

Plan:

1) **Density matrices**
2) **Schmidt decomposition and SVD**
3) **Examples of matrix product states**
4) **Construction of matrix product states**

Density matrix

So far systems were assumed to be in a given quantum state

Classical ignorance about the quantum state

How to express it?

If system is in state $|\psi_i\rangle$ with probability p_i then we should have

$$\langle \mathcal{O} \rangle = \sum_i p_i \langle \psi_i | \mathcal{O} | \psi_i \rangle$$

Superposition won't work: $|\psi\rangle = \sum_i \alpha_i |\psi_i\rangle$

$$\langle \mathcal{O} \rangle \neq \langle \psi | \mathcal{O} | \psi \rangle = \sum_{i,j} \alpha_i \alpha_j \langle \psi_i | \mathcal{O} | \psi_j \rangle$$ **mixed terms**



We need an operator!

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i| \longrightarrow \mathrm{tr}(\hat{\rho}\mathcal{O}) = \sum_i p_i \langle \psi_i | \mathcal{O} | \psi_i \rangle$$



Density matrix

Pure and mixed states

The density matrix obeys few properties:

1) $\operatorname{tr}\hat{\rho} = 1$
2) $\hat{\rho} = \hat{\rho}^\dagger$
3) **semi-positive definite**

→

This implies that eigenvalues $\{\lambda_1, \lambda_2, \dots\}$ of $\hat{\rho}$:

1) **sum up to 1:** $\sum_j \lambda_j = 1$
2) **are real**
3) **are nonnegative**

Two main types of density matrices:

1) **pure:** $\hat{\rho}^2 = \hat{\rho}$
2) **mixed: not pure**

Equivalent characterizations of the pure state:

1) $\hat{\rho} = |\psi\rangle\langle\psi|$
2) $\operatorname{tr}\hat{\rho}^2 = 1$

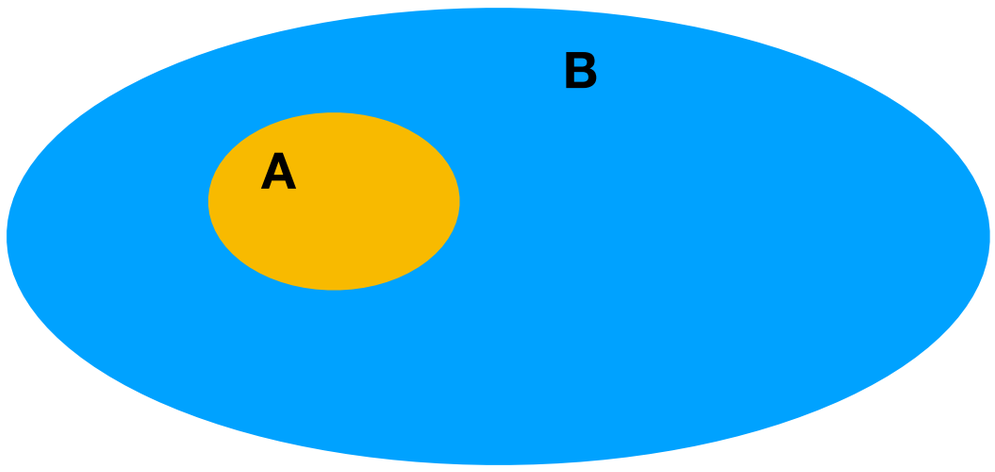
Example:

1) $\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

2) $\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

A pure state is a state which is not a mixture (ensemble) of quantum states

Local observables



whole system = A + B

$$\mathscr{H} = A \otimes B$$

local operators $\mathcal{O}_A$ act only in A

$$\mathcal{O}_A = \mathcal{O} \otimes 1$$

what do we need to know about B to determine

$$\langle\psi|\,\mathcal{O}_A\,|\psi\rangle \text{ ?}$$

If $|\psi\rangle$ is a product state $|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$

$$\langle\psi|\,\mathcal{O}_A\,|\psi\rangle = \langle\psi_A|\,\mathcal{O}\,|\psi_A\rangle \times \langle\psi_B|\psi_B\rangle = \langle\psi_A|\,\mathcal{O}\,|\psi_A\rangle$$

to learn about observables in A
we don't need any information about B

Extract from $|\psi\rangle$ information relevant for observables in A

$$\langle\psi|\,\mathcal{O}_A\,|\psi\rangle = \mathrm{tr}_{A\otimes B}\left(\rho\mathcal{O}_A\right) = \mathrm{tr}_A\left(\mathrm{tr}_B\left(\rho\mathcal{O}_A\right)\right) = \mathrm{tr}_A\left(\mathrm{tr}_B\left(\rho(\mathcal{O}\otimes 1)\right)\right) = \mathrm{tr}_A\left(\mathcal{O}\,\mathrm{tr}_B\left(\rho\right)\right) = \mathrm{tr}_A\left(\mathcal{O}\rho_A\right)$$

Reduced density matrix $\rho_A = \mathrm{tr}_B\,\rho$

Reduced density matrix

Properties of the reduced density matrix ρ_A:

1) **It is a density matrix**
2) **It is pure only if $|\psi\rangle$ is a product state**
3) **ρ_A and ρ_B have the same spectrum (up to the number of zero eigenvalues)**

we will prove 3) later

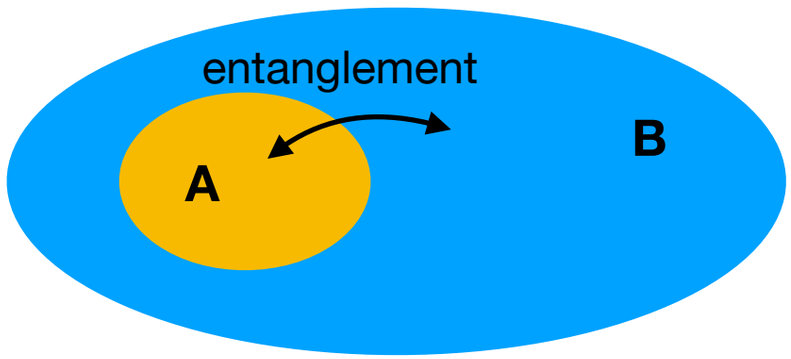
Reduced density matrix ρ_A describes entanglement between A and B:

entanglement entropy - thermodynamic entropy of the reduced density matrix

$$S_A = \mathrm{tr}\,(\rho_A \ln \rho_A)$$

observations: $S_A = S_B$ **- follows from 3)**

$$S_A = 0 \quad \text{for pure } \rho_A$$



entanglement Hamiltonian

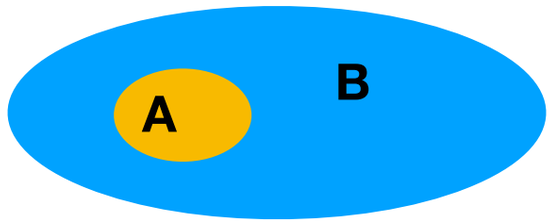
$$\rho_A = e^{-H_{\mathrm{ent}}}$$

Schmidt decomposition

Schmidt decomposition

whole system = A + B

$$\mathscr{H} = A \otimes B$$



We always have

$$|\psi\rangle = \sum_{i,j} C_{i,j} |a_i\rangle \otimes |b_j\rangle$$

however, there is a preferred basis

Schmidt decomposition:

$$|\psi\rangle = \sum_{i=1}^{r} d_i |\alpha_i\rangle \otimes |\beta_i\rangle$$

rank r: number of non-zero d_i

if $r = 1$ then $|\psi\rangle$ is a product state

Example: ground state of XXX

$$|s\rangle = \frac{1}{\sqrt{2}} |\uparrow\downarrow\rangle - \frac{1}{\sqrt{2}} |\downarrow\uparrow\rangle$$

in local spins $a_i, b_j = \{|\uparrow\rangle, |\downarrow\rangle\}$

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Instead

$$\alpha_i = \{|\uparrow\rangle, |\downarrow\rangle\}, \beta_i = \{|\downarrow\rangle, |\uparrow\rangle\}$$

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Entanglement and Schmidt decomposition

Schmidt decomposition:

$$|\psi\rangle = \sum_{i=1}^{r} d_i |\alpha_i\rangle \otimes |\beta_i\rangle$$

Density matrix:

$$\rho = |\psi\rangle\langle\psi| = \sum_{i,j}^{r} d_i d_j^* |\alpha_i\rangle\langle\alpha_j| \otimes |\beta_i\rangle\langle\beta_j|$$

Reduced density matrix:

$$\rho_A = \mathrm{tr}_B \rho$$

$$\rho_A = \mathrm{tr}_B \rho = \mathrm{tr}_B \left(|\psi\rangle\langle\psi| \right) = \sum_{i,j} d_i d_j^* |\alpha_i\rangle\langle\alpha_j| \times \mathrm{tr}_B \left(|\beta_i\rangle\langle\beta_j| \right) = \sum_i |d_i|^2 |\alpha_i\rangle\langle\alpha_j|$$

Entanglement entropy:

$$S_A = \mathrm{tr}_A \left(\rho_A \ln \rho_A \right)$$

$$S_A = \mathrm{tr}_A \left(\rho_A \ln \rho_A \right) = \sum_{i=1}^{r} |d_i|^2 \ln |d_i|^2$$

if we know the Schmidt decomposition

we know the entanglement entropy

Also:

$$\rho_B = \sum_i |d_i|^2 |\beta_i\rangle\langle\beta_j|$$

note that $\rho_A \neq \rho_B$ but their spectra are the same and $S_A = S_B$

(up to zero eigenvalues)

Singular value decomposition

Why does the Schmidt decomposition exists?

$$|\psi\rangle = \sum_{i,j} C_{i,j} |a_i\rangle \otimes |b_j\rangle \xrightarrow{?} |\psi\rangle = \sum_{i=1}^{r} d_i |\alpha_i\rangle \otimes |\beta_i\rangle$$

Singular value decomposition (SVD)

for an arbitrary rectangular ($m \times n$) complex matrix:

$$C = UDV^{\dagger}$$

where:

1) D **is rectangular diagonal matrix of size** $m \times n$ **with nonnegative entries**

$$D = \begin{pmatrix} d_1 & \\ 0 & d_2 \\ 0 & 0 \end{pmatrix}$$

2) U **is unitary square matrix (**$m \times m$**)**

3) V **is unitary square matrix (**$n \times n$**)**

Why does it hold?

$CC^{\dagger}$ **is hermitian and can be diagonalised, the rest (kinda) follows**

Singular value decomposition

How does it work?

1) eigenproblem of $C^\dagger C$

$$C = UDV^\dagger$$

then

$$C^\dagger C = (UDV^\dagger)^\dagger UDV^\dagger = V(D^\dagger D)V^\dagger$$

which gives

$$V^\dagger(C^\dagger C)V = (D^\dagger D)$$

hence the eigenproblem of $C^\dagger C$ gives us V and D

2) unitary U

$$C = UDV^\dagger \rightarrow CV = UD$$

denoting eigenvectors v_i

$$Cv_i = d_i u_i$$

from which we construct U

Singular value decomposition (SVD)

for an arbitrary rectangular ($m \times n$) complex matrix:

$$C = UDV^\dagger$$

where:

1) D **is rectangular diagonal matrix of size** $m \times n$ **with nonnegative entries**
2) U **is unitary square matrix (**$m \times m$**)**
3) V **is unitary square matrix (**$n \times n$**)**

Example: for $C = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$

we get

$$D = \sqrt{5}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

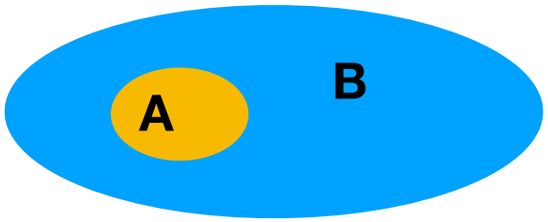
$$U = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$$

$$V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Schmidt decomposition

whole system = A + B

$$\mathscr{H} = A \otimes B$$



We always have

$$|\psi\rangle = \sum_{i,j} C_{i,j} |a_i\rangle \otimes |b_j\rangle$$

Apply SVD to C

$$|\psi\rangle = \sum_{i,j} C_{i,j} |a_i\rangle \otimes |b_j\rangle = \sum_{i,j} (UDV^\dagger)_{ij} |a_i\rangle \otimes |b_j\rangle = \sum_{k=1}^{r} d_k |\alpha_k\rangle \otimes |\beta_k\rangle$$

where $|\alpha_k\rangle = \sum_i U_{ik} |a_i\rangle$, $|\beta_k\rangle = \sum_j V_{jk} |\beta_j\rangle$

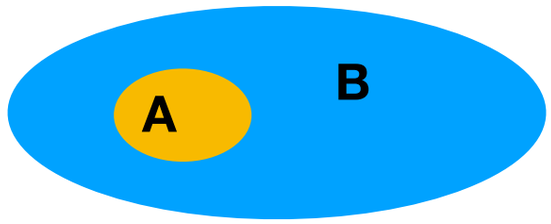
Schmidt decomposition

Entanglement entropy

Entanglement entropy

whole system = A + B

$$\mathscr{H} = A \otimes B$$

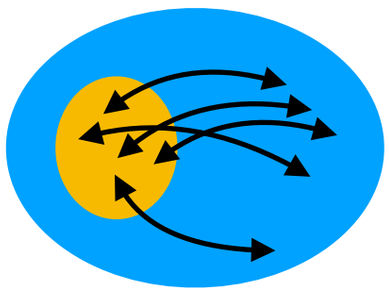


Entanglement entropy as a function of the subsystem size

volume law

$S(A) \sim \mathrm{vol}(A)$

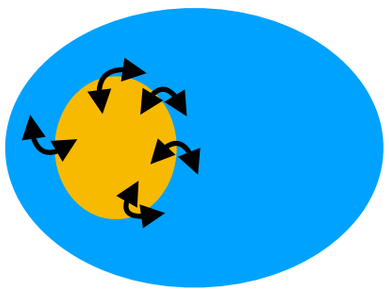
finite temperatures



area law

$S(A) \sim \mathrm{area}(A)$

zero temperatures
gapped phase



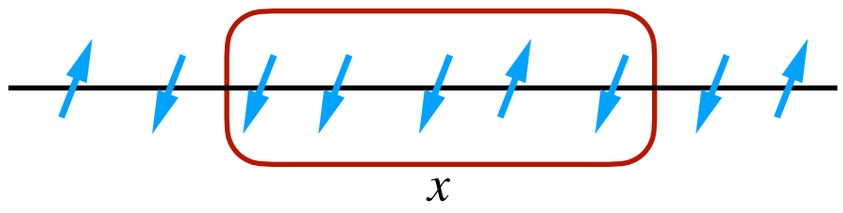
in 1D

$S(A) \sim \log(\xi/a)$

anomalous law

$S(L) \sim c \log(x/a)$

zero temperatures
gapless phase
described by CFT in 1d



Ising chain
XXZ with $|\Delta| < 1$

Matrix Product States

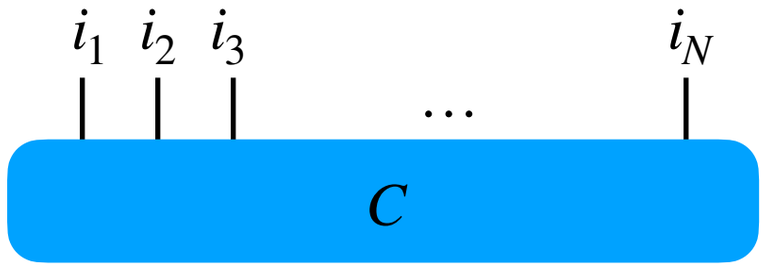
Matrix Product States

Recall tensor product structure of the Hilbert space $\mathscr{H} = \bigotimes_{j=1}^{N} \mathfrak{h}_j$

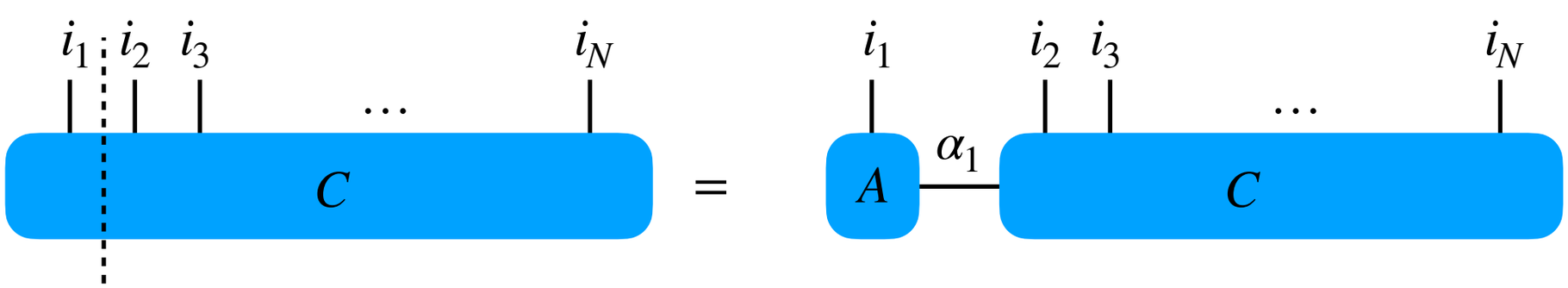
We always have

$$|\psi\rangle = \sum_{i_1,\ldots,i_N} C_{i_1,\ldots,i_N} |a_{i_1}\rangle \otimes \ldots \otimes |a_{i_N}\rangle$$

Large tensor



We can do SVD on it



This gives

$$|\psi\rangle = \sum_{i_1,\ldots,i_N} \sum_{\alpha_1} A^{i_1}_{\alpha_1} C_{\alpha_1,(i_2,\ldots,i_N)} |a_{i_1}\rangle \otimes \ldots \otimes |a_{i_N}\rangle$$

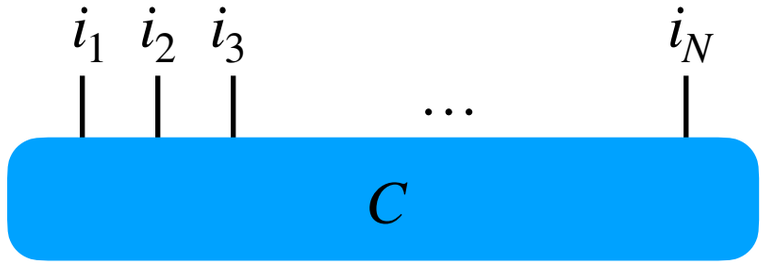
Matrix Product States

Recall tensor product structure of the Hilbert space $\mathscr{H} = \bigotimes_{j=1}^{N} \mathfrak{h}_j$

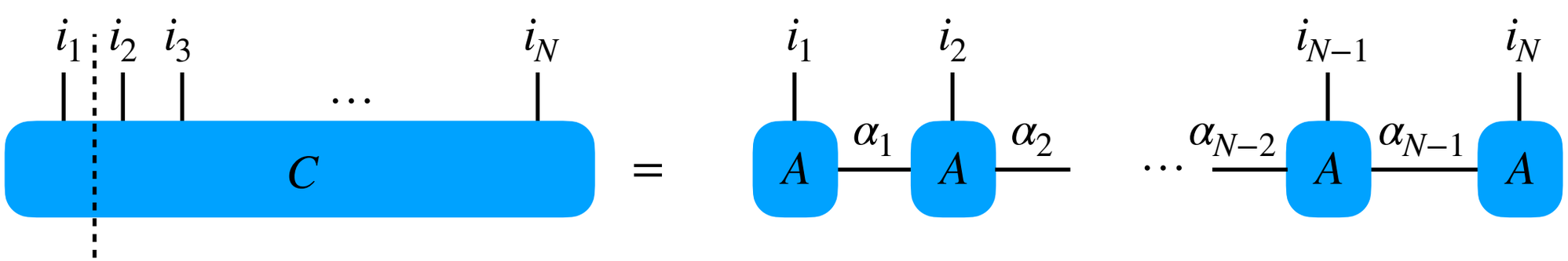
We always have

$$|\psi\rangle = \sum_{i_1,\ldots,i_N} C_{i_1,\ldots,i_N} |a_{i_1}\rangle \otimes \ldots \otimes |a_{i_N}\rangle$$

Large tensor



Iterating



This gives

$$|\psi\rangle = \sum_{i_1,\ldots,i_N} \sum_{\alpha_1,\ldots,\alpha_{N-1}}^{r} A^{i_1}_{\alpha_1} A^{i_2}_{\alpha_1,\alpha_2} \cdots A^{i_{N-1}}_{\alpha_{N-2},\alpha_{N-1}} A^{i_N}_{\alpha_{N-1}} |a_{i_1}\rangle \otimes \ldots \otimes |a_{i_N}\rangle$$

bond dimension r

Exact MPS

Majumdar-Ghosh model

$$H = \sum_{j=1}^{N} \left(\vec{s}_j \cdot \vec{s}_{j+1} + \frac{1}{2} \vec{s}_j \cdot \vec{s}_{j+2} \right)$$

Exact ground state:

$$|\psi\rangle = |s\rangle \otimes |s\rangle \otimes \ldots \otimes |s\rangle \quad \textbf{where} \quad |s\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right)$$

MPS

$$A^{\uparrow} = \begin{pmatrix} 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \quad A^{\downarrow} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 0 & 0 \end{pmatrix}, \quad \textbf{bond dimension = 3}$$